

MACD

Reading Momentum and Trend Shifts — Lesson Explainer

The Concept

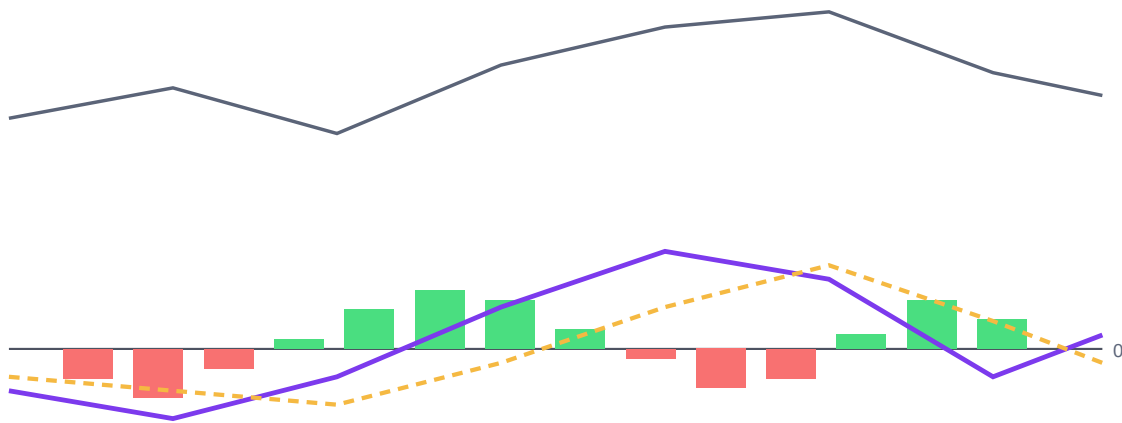
MACD is built from two exponential moving averages (EMAs) of price: a fast 12-period EMA and a slower 26-period EMA. The MACD line is the fast EMA minus the slow EMA.

A 9-period EMA of the MACD line itself is the signal line. When MACD crosses above the signal line, that's read as bullish - momentum picking up. When it crosses below, that's bearish - momentum fading.

The gap between the two is often drawn as a histogram. A growing histogram means the lines are pulling apart; a shrinking one means they're converging - often visible before the crossover itself.

MACD Alongside Price

Price



MACD (purple) / Signal (gold, dashed) / Histogram

5 Things to Know About MACD

1. MACD is a lagging indicator built from moving averages - it confirms a shift after it's begun.
2. The crossover is the headline signal, but the histogram often moves first.
3. MACD can diverge from price, just like RSI - a classic warning sign.
4. The zero line matters too - crossing zero means the fast EMA has overtaken the slow EMA outright.
5. MACD works best in trending markets - in a choppy, sideways market it can whipsaw.

4 Things to Check When Reading MACD

- Watch the histogram shrink - often more useful than reacting to the cross itself.
- Note the zero-line cross - a crossover that also crosses zero carries more weight.
- Check for divergence - compare MACD's recent peaks/troughs to price's.
- Confirm the market isn't choppy - MACD is far more reliable in a trending market.

Worth knowing:

MACD is a lagging indicator built from moving averages - it confirms a momentum shift after it's begun, not before. In choppy, range-bound markets it can generate frequent false crossovers. Pair it with trend context and volume.

Quiz — Answer the questions, then check the key on the next page

1. How is the MACD line calculated?

- A. Today's closing price minus yesterday's closing price
- B. The 12-period EMA minus the 26-period EMA
- C. Total volume divided by average volume
- D. The highest high minus the lowest low

2. What does the MACD histogram represent?

- A. Total trading volume for the day
- B. The company's earnings per share
- C. The gap between the MACD line and the signal line
- D. The stock's dividend yield

3. Why might a shrinking histogram be useful before a crossover?

- A. It never provides any useful information
- B. It guarantees a crossover within 24 hours
- C. It means volume has stopped entirely
- D. It can hint momentum is fading and a crossover may be approaching

4. Why does a crossover that also crosses zero carry more weight?

- A. It means the fast EMA has overtaken the slow EMA outright
- B. Zero-line crosses are purely decorative
- C. It means the stock has been delisted
- D. It only happens once per year

5. Why can MACD generate false signals in a sideways market?

- A. MACD only works on cryptocurrencies
- B. With no sustained trend, the averages cross back and forth without a real shift
- C. The MACD formula breaks down mathematically in sideways markets
- D. Sideways markets have no volume at all

Answer Key

1. How is the MACD line calculated?

Correct answer: B. The 12-period EMA minus the 26-period EMA

The MACD line is the fast (12-period) EMA minus the slow (26-period) EMA of price.

2. What does the MACD histogram represent?

Correct answer: C. The gap between the MACD line and the signal line

The histogram bars plot MACD minus signal - growing means pulling apart, shrinking means converging.

3. Why might a shrinking histogram be useful before a crossover?

Correct answer: D. It can hint momentum is fading and a crossover may be approaching

A shrinking histogram shows the lines converging - often visible before the actual crossover.

4. Why does a crossover that also crosses zero carry more weight?

Correct answer: A. It means the fast EMA has overtaken the slow EMA outright

Crossing zero means the fast EMA has actually moved past the slow EMA, a more decisive shift.

5. Why can MACD generate false signals in a sideways market?

Correct answer: B. With no sustained trend, the averages cross back and forth without a real shift

MACD works best when there's an actual trend to track - a choppy market causes frequent crosses without a genuine shift underway.